Rutgers University
Fall 2021
Math 151 Sections 54-56

Workshop 12

Problem 1

Keywords: Antiderivatives, Initial Value Problem

Let $f(x)$ be a function such that $f''(x) = 8x + 2$ and $y = 10x - 4$ is the tangent line to the graph of f when $x = 1$. Find the function $f(x)$.

Problem 2

Keywords: Areas, Riemann Sums

The goal of this problem is to find formulas for areas under the parabola $y = x^2$.

(a) Using the right endpoint Riemann sum with 4 intervals of equal length, approximate the area under the parabola $y = x^2$ between $x = 0$ and $x = 2$, as well as the average value of $y = x^2$ on $[0, 2]$.

(b) Let S_n be the right endpoint Riemann sum with n intervals of equal length for the function $y = x^2$ on the interval $[0, b]$, where b is a positive constant. Find an expression for S_n using the Σ symbol.

(c) Using the fact that

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6},$$

write a simplified expression of S_n which does not involve the Σ symbol.

(d) Deduce an expression for $\displaystyle\int_0^b x^2 \mathrm{d}x$, the area under the parabola between $x = 0$ and $x = b$.

(e) Using your previous result, find an expression for $\displaystyle\int_a^b x^2 \mathrm{d}x$, where $0 < a < b$ are two constants.

(f) Find an expression for $\displaystyle\int_{-b}^{-a} x^2 \mathrm{d}x$, where $0 < a < b$ are two constants. (Hint: graph the function and compare the areas under the curve on the intervals $[a, b]$ and $[-b, -a]$.)

Problem 3

Keywords: Limits of Riemann Sums, Definite Integrals, Properties of the Integral

Let f be a continuous function such that

$$\int_1^4 f(x)\mathrm{d}x = 6.$$

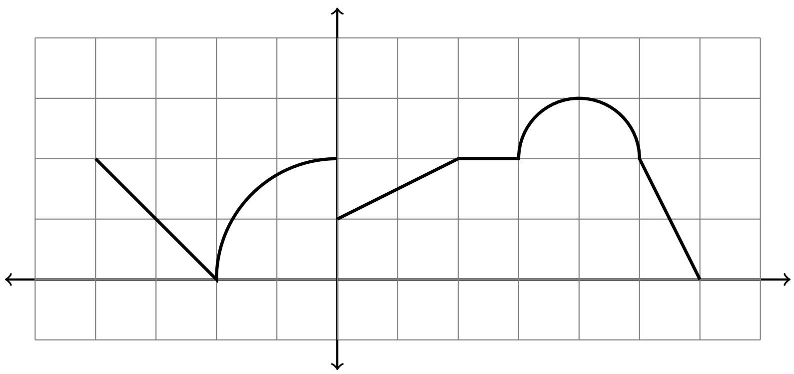
(a) The limit of the following Riemann sum representes the integral of a certain function $g(x)$ from $x = a$ to $x = b$:

$$\lim_{n\to\infty} \sum_{k=1}^{n} \left(2f\left(1 + \frac{3k}{n}\right) + \left(1 + \frac{3k}{n}\right)^2 \right) \frac{3}{n} = \int_a^b g(x)\mathrm{d}x.$$

Identify a suitable function $g(x)$ and bounds a, b.

(b) For your choice of $g(x)$, a and b, evaluate $\int_a^b g(x)\mathrm{d}x$. You may have to use your results from the previous problem.

Extra Problem *(for practice, no need to submit)*
Let $f(x)$ be the function graphed below.



- The part above $[-2, 0]$ is a quarter-circle with center (0,0) and radius 2.
- The part above $[3, 5]$ is a semi-circle with center (4,2) and radius 1.

(a) Compute $\int_0^6 f(x)\mathrm{d}x$ and the average value of f on $[0, 6]$.

(b) Find two values of the pair (a, b) for which $\int_a^b f(x)\mathrm{d}x = \pi + 5$.